

ShopSphere

S5 - Inventory, Reservation and Stock Control

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

1. Sprint Definition

Sprint: S5 - Inventory, Reservation and Stock Control

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Inventory, Reservation and Stock Control, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

2. Business Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01	Reserve stock	Checkout service	SKU has available inventory	Lock/validate, increase reserved stock, decrease available stock	Reservation succeeds atomically.
UC-02	Adjust stock	Inventory admin	Authorized role	Enter adjustment and reason	Stock changes and immutable transaction record is created.

3. Detailed Implementation Plan

1. Implement total/reserved/available stock accounting.
2. Implement stock adjustment with transaction history.
3. Implement reserve/release/consume operations using safe transactions/locking.
4. Prevent negative available stock and overselling.

4. Data / State / Contract Scope

- inventory
- inventory_transactions

API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

7. Test Cases

TC	Scenario	Expected Result
TC-S5-01	Zero stock	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-02	Negative adjustment	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-03	Reserve] available	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-04	Concurrent reservation	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-05	Unknown release	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-06	Duplicate adjustment	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-07	Inactive SKU	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S5-08	Transaction rollback	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S5 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S5 document for that platform.